

NPTEL
NPTEL ONLINE CERTIFICATION COURSE

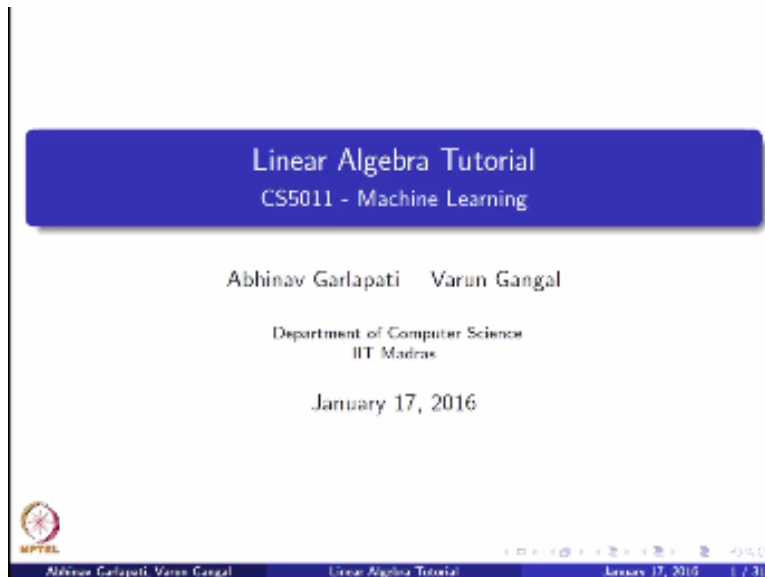
Reinforcement Learning

Week: 0

Lecture: 3

Tutorial - Linear Algebra 2
Prof. Balaraman Ravindran
Department of Computer Science and
Engineering
Indian Institute of Technology Madras.

(Refer Slide Time: 00:15)



The slide features a blue header bar with the text "Linear Algebra Tutorial" and "CS5011 - Machine Learning" in white. Below the header, the names "Abhinav Garlapati" and "Varun Gangal" are listed, followed by "Department of Computer Science" and "IIT Madras". The date "January 17, 2016" is centered at the bottom. A small NPTEL logo is in the bottom left corner, and a navigation bar with icons and the text "Abhinav Garlapati, Varun Gangal", "Linear Algebra Tutorial", "January 17, 2016", and "(/ 31)" is at the very bottom.

Linear Algebra Tutorial
CS5011 - Machine Learning

Abhinav Garlapati Varun Gangal

Department of Computer Science
IIT Madras

January 17, 2016

NPTEL

Abhinav Garlapati, Varun Gangal Linear Algebra Tutorial January 17, 2016 (/ 31)

Hi, everyone welcome to the second tutorial of this, of the introduction to machine learning course, so in this tutorial we shall be taking a tour of the aspects of linear algebra which you would need for the course, we will cover a variety of concepts, such as subspaces, basis, span and decompositions, Eigen values, Eigen vectors over the course of the tutorial.

(Refer Slide Time: 00:47)

What is Linear Algebra

Linear Algebra

Linear algebra is the branch of mathematics concerning vector spaces and linear mappings between such spaces. It includes the study of lines, planes, and subspaces, but is also concerned with properties common to all vector spaces.

Why do we study Linear Algebra?

- Provides a way to compactly represent & operate on sets of linear equations.
- In machine learning, we represent data as matrices and hence it is natural to use notions and formalisms developed in Linear Algebra.

 NPTEL

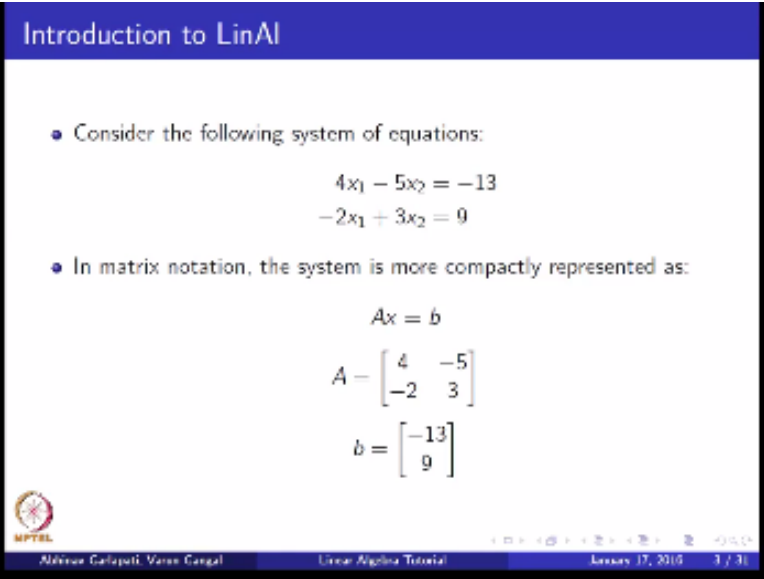
Aditya Garg, IIT Bombay Linear Algebra Tutorial January 17, 2019 2 / 36

So the first question one would ask is why we need linear algebra at all and what is linear algebra, so you may have come across this in school or your + 1 or + 2 level but just to recap I mean linear algebra is the branch of mathematics which deals with vectors and vector spaces and the linear mappings between these spaces. So why do we study linear algebra here especially in the context of machine learning is firstly it gives us a way to manipulate, represent and operate sets of linear equations.

But why do these linear equations come up in machine learning in the first place. So the reason for this is that in machine learning we represent our data in an n cross d matrix where n is in this number of data points and p is the number of features. So it is natural that we have to use notions and formalisms developed in linear algebra. Not just the data even I mean the parameters you

use are represented as vectors so as a result linear algebra has an important role to play in machine learning.

(Refer Slide Time: 02:05)



The slide is titled "Introduction to LinAl" in a blue header. It contains two bullet points. The first bullet point says "Consider the following system of equations:" followed by two equations: $4x_1 - 5x_2 = -13$ and $-2x_1 + 3x_2 = 9$. The second bullet point says "In matrix notation, the system is more compactly represented as:" followed by the matrix equation $Ax = b$. Below this, matrix A is defined as $A = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix}$ and vector b is defined as $b = \begin{bmatrix} -13 \\ 9 \end{bmatrix}$. At the bottom left is the NPTEL logo. At the bottom right is a navigation bar with icons and the text "Linear Algebra Tutorial" and "Session 17, 2016 1 / 31".

Introduction to LinAl

- Consider the following system of equations:
$$4x_1 - 5x_2 = -13$$
$$-2x_1 + 3x_2 = 9$$
- In matrix notation, the system is more compactly represented as:
$$Ax = b$$
$$A = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix}$$
$$b = \begin{bmatrix} -13 \\ 9 \end{bmatrix}$$

NPTEL

Linear Algebra Tutorial Session 17, 2016 1 / 31

So we can see here a system of linear equations two equations and two variables

$$4x_1 - 5x_2 = -13$$

and

$$-2x_1 + 3x_2 = 9$$

so we can right away see here the advantages of the matrix notation so if you see below you can represent the same system of two equations directly as one equation in the form of Ax equals b , where A is the set of coefficient and x is the 2×1 matrix, 2 rows and 1 column or you may also call it a two dimensional vector x_1, x_2 so you can see that when you multiply the matrix A with x_1, x_2 so you can see that when you multiply the matrix A with x_1, x_2 the 2×1 matrix you will get back the same LHS or the set of two LHSES and b the matrix b represents the RHS.

So we can it is very easy to verify that if you represent different matrices you can get back the original representation and from that representation you can get this, so solving this in general without matrices how you would solve this is you would first solve the one variable and then substitute to get the other but this using matrices you can even solve this directly so just multiply both the sides by A^{-1} , so you would get x and $A^{-1}b$ of course the nuance you would want to care about is that all matrices do not have an inverse but in most cases they do.

So in that case you can directly get the solution of x in the form of x equals $A^{-1}b$ by solving it just through one equation. So as we said earlier linear algebra gives you this freedom to manipulate several equations at once in multiple variables.

(Refer Slide Time: 04:09)

Vector Space

Definition

A set V with two operations $+$ and $\cdot$ is said to be a **vector space** if it is closed under both these operations and satisfies the following eight axioms.

1 Commutative Law

$$x + y = y + x, \quad \forall x, y \in V$$

2 Associative Law

$$(x + y) + z = x + (y + z), \quad \forall x, y, z \in V$$

3 Additive identity

$$\exists 0 \in V \text{ s.t. } x + 0 = x, \quad \forall x \in V$$

4 Additive inverse

$$\forall x \in V, \exists \tilde{x} \text{ s.t. } x + \tilde{x} = 0$$



Mahesh Gopalatti, Vares Gargal
Linear Algebra Tutorial
January 16, 2016
4 / 31

A fundamental definition in linear algebra is that of a vector space, a set of vectors V is said to be a vector space if it is closed under the operations of vector addition and scalar multiplication and in addition satisfies the axioms we have listed here.

By closed, we mean that if we take two elements from this set x and y then $x + y$ it also lies in the set V in addition to this if we take a scalar α or real number that is and multiply a vector from this set y with it then αy also belongs to V . If both these properties are satisfied then the set of vectors is said to be closed with respect to vector addition and scalar multiplication.

Now let us have a look at the axioms the first one is the commutative law, the commutative law states that if you pick any two elements from the set V , x and y then $x + y = y + x$. The Associative law says that if you pick any three element from this set x , y , z then $x + y$ added together $+ z = x + y + z$ added together. The additive identity law says that there exists an additive identity or 0 so as to say in this set.

Such that if you pick any element from the set and add the 0 to it you get back the same element. The additive inverse law says that there exists for every element there exists another for every element x there exists a corresponding $\bar{x}$ such that $x + \bar{x}$ equals 0.

(Refer Slide Time: 06:15)

Vector Space (Contd..)

- Distributive Law
$$\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y, \quad \forall \alpha \in \mathbb{R}, x, y \in V$$
- Distributive Law
$$(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x, \quad \forall \alpha, \beta \in \mathbb{R}, x \in V$$
- Associative Law
$$(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x), \quad \forall \alpha, \beta \in \mathbb{R}, x \in V$$
- Unitary Law
$$1 \cdot x = x, \quad \forall x \in V$$



MPPEL

Alfonso Galuppi, Vane Gargal

Linear Algebra Tutorial

January 10, 2015

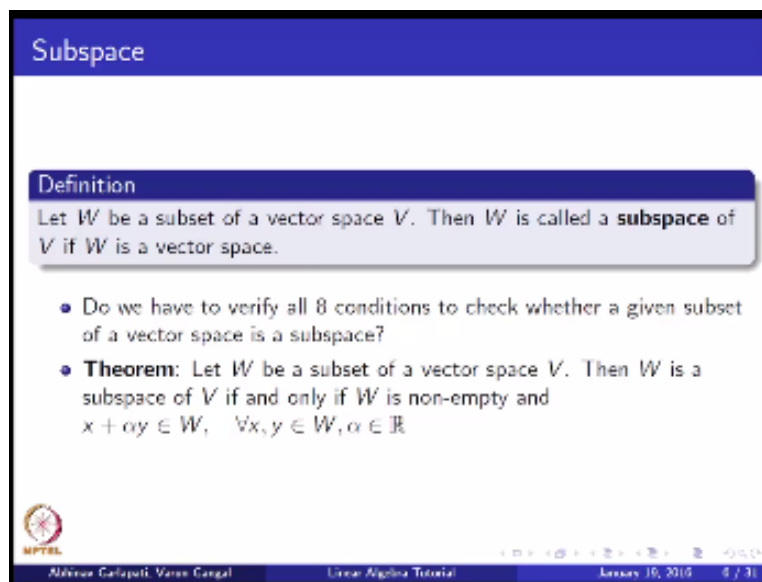
5 / 31

The fifth law is the distributive law, this law says that if you have a scalar α with which you multiply the sum of two vectors $x + y$ then that should be equal to $\alpha x + \alpha y$. The second

distributive law says that if you have the sum of two scalars multiplying a vector x from the set $(\alpha + \beta)x$ then that should be equal to $\alpha x + \beta x$. The associative law says that if you first multiply two scalars and then multiply the vector with them that should be equal to multiplying it first with second scalar β and then with α .

The unitary law says that on multiplication by this scalar real number 1 you get back the same vector, so this is important because we don't want multiplication to pose any unexpected scaling, if you multiply by the scalar k then the vector you get should be exactly k times it should not be say root k .

(Refer Slide Time: 07:34)



Subspace

Definition
Let W be a subset of a vector space V . Then W is called a **subspace** of V if W is a vector space.

- Do we have to verify all 8 conditions to check whether a given subset of a vector space is a subspace?
- **Theorem:** Let W be a subset of a vector space V . Then W is a subspace of V if and only if W is non-empty and $x + \alpha y \in W, \forall x, y \in W, \alpha \in \mathbb{R}$

 NPTEL
 Abhinav Garg, IIT Bombay
 Linear Algebra Tutorial
 January 16, 2016 6 / 31

A second related definition is that of a subspace, a subset W of a vector space V is said to be a subspace in W is a vector space, now this means that W should be closed under vector addition and scalar multiplication, it should also satisfy the eight axioms we stated earlier. Now the question that arises is do we need to verify all these eight conditions given that we know that W is already a subset of a vector space.

It is enough to check just for the following two conditions, firstly that W has to be non empty that in other words it has at least a single element and secondly that if I pick any two elements x and y from this set and any real number α then $x + \alpha y$ should belong to W .

(Refer Slide Time: 08:39)

Norm

Definition

Norm is any function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying:

- 1 $\forall x \in \mathbb{R}^n, f(x) \geq 0$ (non-negativity)
- 2 $f(x) = 0$ iff $x = 0$ (definiteness)
- 3 $\forall x \in \mathbb{R}^n, f(tx) = |t|f(x)$ (homogeneity)
- 4 $\forall x, y \in \mathbb{R}^n, f(x + y) \leq f(x) + f(y)$ (triangle inequality)

- Example - l_p norm

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$
- Matrices can have norms too - e.g., Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(A^T A)} \quad (1)$$



Aditya Garg, Vignesh Garg
Linear Algebra Tutorial
January 10, 2019
7 / 31

Now let us have look at a definition of a norm, so intuitively a norm is a measure of the length of a vector or its magnitude, it is a function from this vector space which mostly happens to be $\mathbb{R}^n$ where n is the dimension of a vector to this space of real numbers $\mathbb{R}$, so for a function to be a norm it should satisfy the four conditions which we have given here. Firstly it should be it should always be non negative.

Secondly, it should be 0 if and only if the vector is 0, thirdly for every vector if you multiply it by a scalar its norm should get multiplied by the modulus of the scalar by modulus here we mean the absolute value and fourth being that if we take any pair of vectors in our vector space which is $\mathbb{R}^n$, the norm of the sum of these two vectors should be less than the sum of their norms this is also known as triangle inequality.

So this is in a way related to the fact that the third side of a triangle should always be less than sum of the other two sides. Now an example of a norm is the L_p norm where you sum up the absolute values along these dimensions

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$

raised to p and then take the 1 by p^{th} root of this, so when p equals 2 you get the l_2 two norm which is the magnitude of a vector that we have learned in our earlier studies $\sqrt{x^2 + y^2}$ if you are looking at just at the space $\mathbb{R}^2$.

There are other norms for instance there are norms defined for matrices as well, here we have defined norms for vectors so the Frobenius norm is a matrix norm

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

so what it does is it essentially sums up the square of all the elements and then takes the root of that, so this also happens to be equal to the trace of A transpose A , now the trace of a matrix is simply the sum of its diagonal elements.

(Refer Slide Time: 11:12)

Range Of A Matrix

- The **span** of a set of vectors $X = \{x_1, x_2, \dots, x_n\}$ is the set of all vectors that can be expressed as a linear combination of the vectors in X .
In other words, set of all vectors v such that $v = \sum_{i=1}^{|X|} \alpha_i x_i, \alpha_i \in R$
- The **range** or **columnspace** of a matrix A , denoted by $R(A)$ is the span of its columns. In other words, it contains all linear combinations of the columns of A . For instance, the column space of $A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix}$ is the plane spanned by the vectors $\begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$



The span of a set of vectors is the set of all vectors which can be composed using these vectors by using the operations of vector addition and scalar multiplication so the name span comes from the fact that this set of vectors spans potentially a larger set of vectors which is then called a span

so to define this more formally it is the set of all vectors v such that $v = \sum_{i=1}^{|X|} \alpha_i x_i, \alpha_i \in R$. Now a related definition is that of range or column space so if we think of a matrix each of its columns is a vector so the set of all columns of a matrix is like a set of vectors.

Now the span of this set of vectors is called the range or column space of that matrix so if you consider the matrix given here the columns of this matrix A are

$$\begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix}$$

so what would be this column space of this matrix it would be the span of the vectors $\begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$ which is essentially the plane which is spanned by these two vectors.

(Refer Slide Time: 12:56)

Nullspace Of A Matrix

Definition
The nullspace $N(A)$ of a matrix $A \in \mathbb{R}^{m \times n}$ is the set of all vectors that equal 0 when multiplied by A . The dimensionality of the nullspace is also referred to as the **nullity** of A .

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

- Note that vectors in $N(A)$ are of dimension n , while those in $R(A)$ are of size m , so vectors in $R(A^T)$ and $N(A)$ are both of dimension n .

 NPTEL
Abhinav Chakraborty, Venkatesh Gopal
Linear Algebra Tutorial
January 19, 2016 9 / 31

If we have a matrix A of dimensions $N \times M$ then the null space is the set of $N \times 1$ vectors which given $M \times 1$ 0 vector on V multiplied by A in other words it is set of vectors X such that $A \cdot X$ equals 0. Nullity is the rank or dimensionality of the null space we will revisit the definition of nullity later once you defined rank more clearly. Another interesting fact about null spaces is that the null space of A , $N(A)$ is of dimension N while the range of A or the columns space as we defined it earlier is of dimension M or $M \times 1$ this means that vectors in R of A^T so note that A^T is $N \times M$.

So vectors in the range of A^T will be of dimension N similar to the vectors in the null space of A so this means that the vectors in the range of A^T and null space of A would both be of the dimension $N \times 1$.

(Refer Slide Time: 14:15)

Another Example

Now, consider the matrix

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix}$$

Here, the third column is a linear combination of the first two columns.
Here, the nullspace is the line of all points $x = c, y = c, z = -c$.



Arjun Karan, Manoj Ganguli Linear Algebra Tutorial January 19, 2016 11 / 31

Let us consider an example to illustrate the concept of a null space consider the matrix A given in here,

$$A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix}$$

A is a 3 X 2 matrix since the null space of A then may be vectors of dimension 2 or 2 X 1 now we see that on solving we get U equals 0 V equals 0 this means that a null space only contains the 0 vector the two dimensional zero vectors 0 , 0 let us consider an another example to get illustrate null spaces better. Take the matrix B which is a 3 X 3 matrix the null space would consist of 3 X 1 vectors. We leave the finding of the null space to the audience as an exercise.

However when on solving we get the null space to be the set of all vectors of the form

$$x = c, y = c, z = -c$$

where c is any real number and x y c refer to the first second and third dimension respectively.

(Refer Slide Time: 15:34)

Linear Independence and Rank

Definition
A set of vectors $\{x_1, x_2, \dots, x_n\} \in \mathbb{R}^n$ is said to be **(linearly) independent** if no vector can be represented as a linear combination of the remaining vectors.

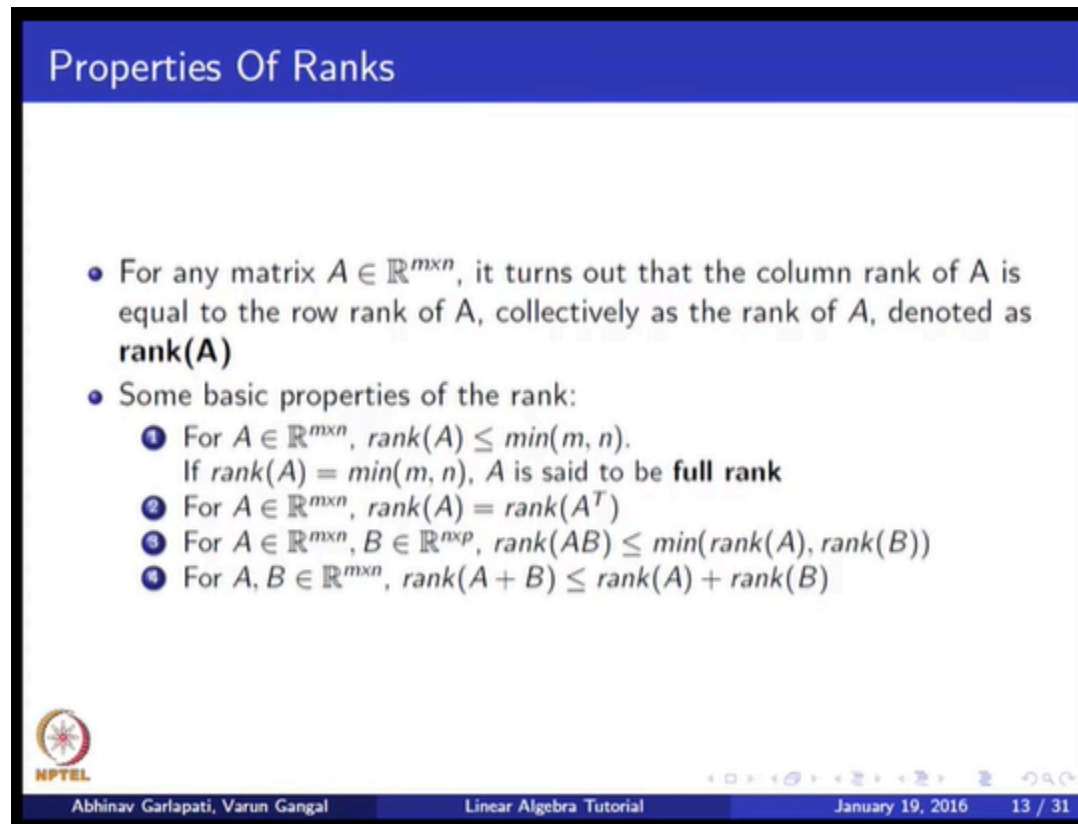
- i.e., if $x_n = \sum_{i=1}^{n-1} \alpha_i x_i$ for some scalar values $\alpha_1, \dots, \alpha_{n-1} \in \mathbb{R}$, then we say that the vectors $\{x_1, x_2, \dots, x_n\}$ are linearly dependent; otherwise, the vectors are linearly independent
- The **column rank** of a matrix $A \in \mathbb{R}^{m \times n}$ is the size of the largest subset of columns of A that constitute a linearly independent set
- Similarly, **row rank** of a matrix is the largest number of rows of A that constitute a linearly independent set

Anshul Garg, Varun Garg | Linear Algebra Tutorial | January 19, 2016 | 12 / 31

Before we dive into defining linear independents recollect how we defined a linear combination a set of vectors is linearly independent if no vector in the set can be produced using a linear combination of the other vectors in this set now let us have a look at related concept of rank so the column rank of a $N \times N$ matrix A is the size of the largest linearly independent subset of columns note that our columns here are $N \times 1$ vectors the row rank is defined in a similar way for rows.

(Refer Slide Time: 16:22)



Properties Of Ranks

- For any matrix $A \in \mathbb{R}^{m \times n}$, it turns out that the column rank of A is equal to the row rank of A , collectively as the rank of A , denoted as **rank(A)**
- Some basic properties of the rank:
 - ① For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) \leq \min(m, n)$.
If $\text{rank}(A) = \min(m, n)$, A is said to be **full rank**
 - ② For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = \text{rank}(A^T)$
 - ③ For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
 - ④ For $A, B \in \mathbb{R}^{m \times n}$, $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

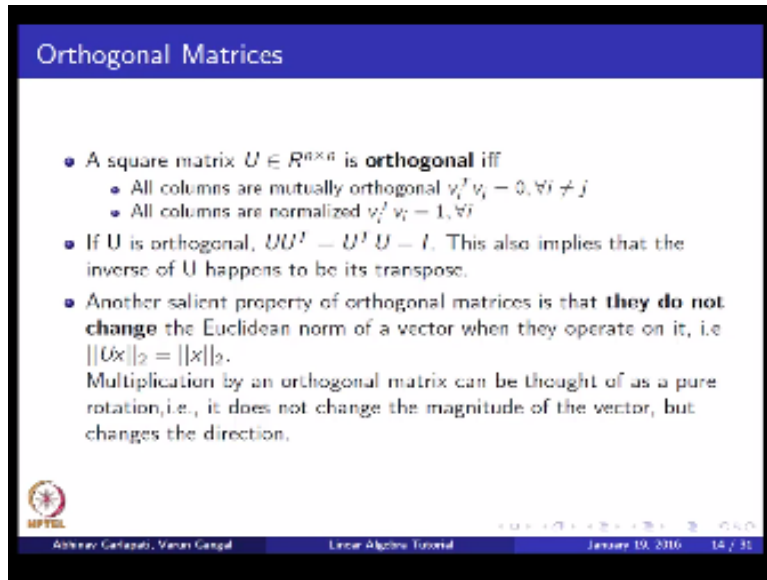
NPTEL

Abhinav Garlapati, Varun Gangal Linear Algebra Tutorial January 19, 2016 13 / 31

Let us walk through some interesting properties of ranks for any $M \times N$ matrix of real numbers. The column rank is equal to the row rank we refer to this quantity as the rank of the matrix earlier we had looked at a quantity called nullity, nullity is the rank of the null space of A . Some other interesting properties of ranks are also listed here. For instance, the rank of a matrix is at most the minimum of its two dimensions, the row dimension and the column dimension. Secondly the rank of a matrix is the same as the rank of its transpose. Thirdly if you multiply two matrices A and P the rank of the resultant matrix is at most the minimum.

Of the ranks of A and B if you add up to matrices the rank of the resultant matrix is at most the sum of the ranks of A and B .

(Refer slide time: 17:34)



Orthogonal Matrices

- A square matrix $U \in \mathbb{R}^{n \times n}$ is **orthogonal** iff
 - All columns are mutually orthogonal $v_i^T v_j = 0, \forall i \neq j$
 - All columns are normalized $v_i^T v_i = 1, \forall i$
- If U is orthogonal, $UU^T = U^T U = I$. This also implies that the inverse of U happens to be its transpose.
- Another salient property of orthogonal matrices is that **they do not change** the Euclidean norm of a vector when they operate on it, i.e. $\|Ux\|_2 = \|x\|_2$.
Multiplication by an orthogonal matrix can be thought of as a pure rotation, i.e., it does not change the magnitude of the vector, but changes the direction.

MIT OCW
Arjun Gupte, Varun Gargal
Linear Algebra, Tutorial
January 19, 2016
14 / 31

A square matrix U of dimension $N \times N$ is defined to be orthogonal if and only if following two conditions hold. Firstly, all pairs of distinct columns should be orthogonal, by columns being orthogonal we mean that the dot product of any pair of distinct column vectors is 0 in other words $V_i^T V_j = 0$ for all i not equal to j . The second condition is that a dot product of any column with itself or $V_i^T V_i$ equals 1.

In other words, all the column vectors should be normalized. An interesting implication of a matrix being orthogonal is that UU^T and $U^T U$ both end up being equal to the $N \times N$ identity matrix I . This also means that $U^T = U^{-1}$ or the transpose of such an orthogonal matrix is also its inverse. An additional interesting property is seen when we multiply a $N \times 1$ vector x by a $N \times N$ orthogonal matrix U , the Euclidean or l_2 norm of such a vector X remains the same on multiplication by U .

Intuitively, we can understand this as orthogonal matrices U performing good pure rotation or multiplying the vector X in other words they only change the direction of a vector but they won't change its magnitude.

(Refer Slide Time: 19:27)

Quadratic Form of Matrices

- Given a square matrix $A \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, the scalar value $x^T A x$ is called a **quadratic form**
- A symmetric matrix $A \in \mathbb{R}^n$ is positive definite (PD) if for all non-zero vectors $x \in \mathbb{R}^n$, $x^T A x > 0$
- Similarly, positive semidefinite if $x^T A x \geq 0$, negative definite if $x^T A x < 0$ and negative semidefinite if $x^T A x \leq 0$
- One important property of positive definite and negative definite matrices is that they are always full rank, and hence, invertible.
- **Gram matrix:** Given any matrix $A \in \mathbb{R}^{m \times n}$, matrix $G = A^T A$ is always positive semidefinite.
Further if $m \geq n$, then G is positive definite.



Arshad Ganiyil, Varun Gangal
Linear Algebra Tutorial
January 19, 2016
15 / 35

We often encounter the quadratic form which is the vector equivalent of a quadratic function the quadratic form with respect to the matrix A of a vector X where the matrix A is $N \times N$ and the vector X is $N \times 1$ is given by the real number $x^T A x$. Based on the quadratic form of the matrices we can classify them as positive definite, negative definite, positive semi definite and negative semi definite. A matrix A is said to be positive definite if its quadratic form is greater than 0 for any vector X . Similarly we can define it to be negative definite. A matrix A is positive semi definite

If the quadratic form is greater than equal to 0 for any vector x . Note that equality with 0 also may hold here but important property of positive and negative definite matrices is that they are always full rank. An implication of this is that A^{-1} always exists for a matrix A which is of dimension $M \times N$. One can define a special matrix called a gram matrix, the gram matrix is given by $A^T A$. The salient property of the gram matrix is that it is always positive semi definite moreover if the number of rows exceeds the number of columns in another words if M is greater than equal to N this means this implies that G is positive definite.

IIT Madras Production

Funded by

Department of Higher Education
Ministry of Human Resource Development

Government of India

www.nptel.ac.in

Copyrights Reserved